

Title of the Project : Sign Rhythm - Smart Glove for Sign Language Translation and Gesture-Controlled Home Automation

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Abstract: The “Smart Glove for Sign Language Translation and Gesture Controlled Home Automation” is an innovative wearable system designed to bridge the communication gap between speech-impaired individuals and the general public while also enabling intuitive control of home appliances. The smart glove integrates an ADXL345 accelerometer sensor to detect and interpret hand gestures in real-time. These gestures are processed by an Arduino microcontroller, which maps them to predefined sign language commands. Once a gesture is recognized, the corresponding message is displayed on an I2C LCD screen and simultaneously converted into speech using a DF Player Mini module and speaker, allowing effective verbal communication. In addition to communication support, the system also enables gesture-based control of household appliances such as lights and fans through relay modules, promoting hands-free automation. The proposed system offers a cost-effective, portable, and user-friendly solution that enhances accessibility and improves the quality of life for differently-abled individuals. By combining assistive communication and home automation in a single device, the project demonstrates a practical approach to inclusive smart living and human-computer interaction.

Project Details: The “Smart Glove for Sign Language Translation and Gesture Controlled Home Automation” is a wearable assistive system that combines gesture recognition with basic home automation. The glove is fitted with an ADXL345 accelerometer sensor that detects hand movements and orientation in three axes. These motion signals are continuously sent to an Arduino microcontroller, which processes the data and compares it with predefined gesture patterns stored in its memory. When a gesture matches a stored pattern, the system performs a specific action. For communication purposes, the translated sign is displayed as text on an I2C LCD screen and converted into speech using a DF Player Mini audio module and speaker, allowing the user’s message to be clearly understood by others. For automation purposes, selected gestures are assigned to control appliances such as lights and fans. The Arduino activates relay modules to switch these devices ON or OFF accordingly. The system operates in real time, providing fast response and reliable performance. Its compact and lightweight design makes it comfortable to wear and easy to use. The integration of communication assistance and gesture-based automation in a single device makes this project an effective and affordable solution for improving accessibility and supporting independent living in smart home environments.

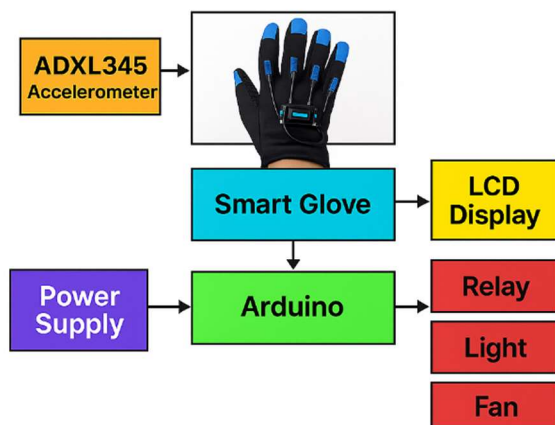


Fig : System Architecture of Sign Rhythm

Result:



Fig : Sign Rhythm

Conclusion: The project successfully demonstrated a wearable smart glove that recognizes hand gestures for both communication and home automation. It allows differently-abled individuals to convey messages and control appliances like lights and fans with ease. The system proved accurate, responsive, and user-friendly, showing strong potential as an affordable and portable solution to improve accessibility and independent living in smart homes.

